

Galaxy Telecom AI Email Triage Agent

An end-to-end AI email triage pipeline on Microsoft Azure — built to demonstrate agentic AI orchestration, secure serverless architecture and production-grade Azure design patterns.

github.com/gayatrianne/email-triage-agent

POC Complete — 24 Jul 2026

Independent upskilling project. Galaxy Telecom is a fictional company built to give the architecture a realistic business context — the Azure resources, code and test evidence below are real. Full source, docs and commit history are on [GitHub](#).

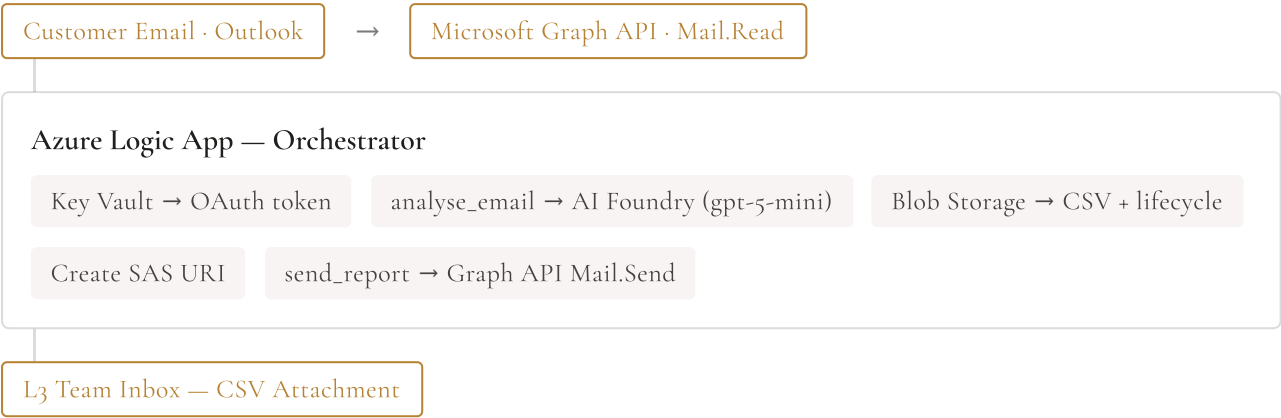
01 — HYPOTHESIS & PROBLEM

"Can an AI agent automatically read customer emails, determine intent and sentiment, and route them to the correct support team — reducing manual triage effort?"

Galaxy Telecom's customer service team spends significant time manually reading, categorising and routing incoming customer emails. This POC tests whether a serverless, AI-driven pipeline can remove that manual step entirely — reading unread emails, classifying intent/sentiment/priority with an LLM, and delivering a structured triage report to L3 support with no human intervention.

02 — WHAT WAS BUILT

An end-to-end pipeline that reads unread helpdesk emails, analyses each with an LLM for intent/sentiment/routing/priority, compiles the results into a CSV, and emails it to the L3 team daily — fully orchestrated, no code in the critical path.



03 — AZURE COMPONENTS

COMPONENT	RESOURCE	PURPOSE
Azure AI Foundry	proj-email-triage-dev	Hosts gpt-5-mini model
Azure Function App	func-email-triage-dev	analyse_email + send_report functions
Azure Logic App	logic-email-triage-dev	Pipeline orchestration
Azure Blob Storage	email-triage-output	CSV storage with lifecycle management
Azure Key Vault	kv-email-triage-dev	OAuth refresh token storage
Microsoft Graph API	app-email-triage-graph	Mail.Read + Mail.Send permissions

04 — TECH STACK

Python 3.11

Azure Functions v2 · Flex Consumption

Azure AI Foundry · gpt-5-mini · Responses API

Azure Logic Apps · Consumption, Stateful

Microsoft Graph API · OAuth 2.0

Azure Key Vault · Managed Identity

Azure Blob Storage · Lifecycle Policy

GitHub Actions CI/CD

05 — AI ANALYSIS OUTPUT

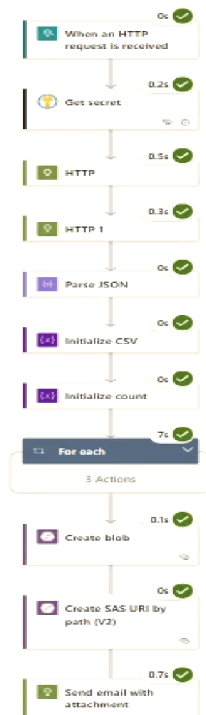
FIELD	VALUES
Intent	billing_query, technical_support, account_management, complaint, general_enquiry
Sentiment	positive, neutral, negative, urgent
Routing Team	Billing, Technical, Account, Complaints, General
Priority	high, medium, low
Summary	One-sentence description of the email

06 — TEST RESULTS

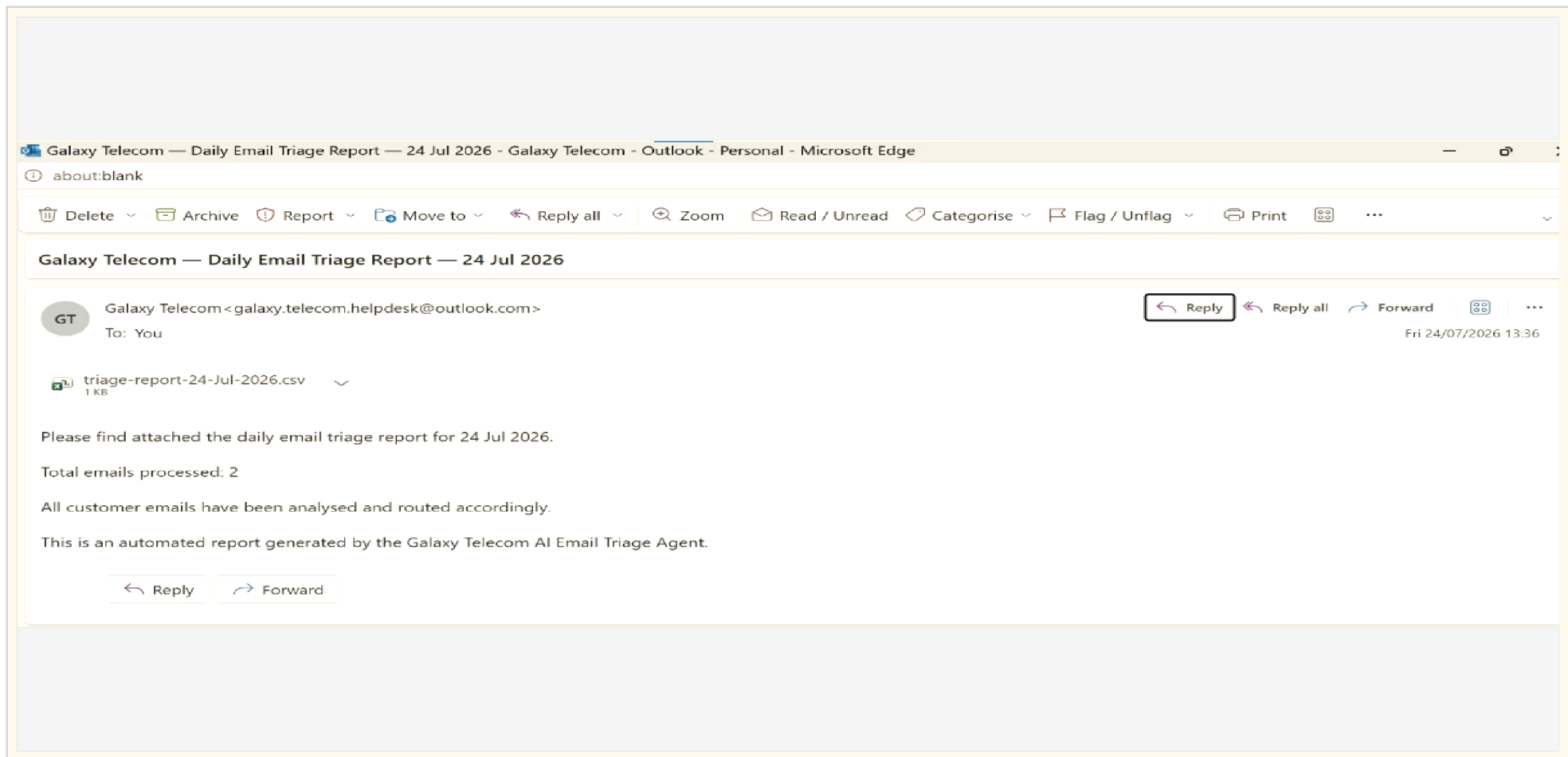
SUBJECT	INTENT	SENTIMENT	TEAM	PRIORITY	RESULT
Mobile Internet Not Working Since Yesterday	technical_support	urgent	Technical	high	✓ Correct
Charged Twice for My Monthly Bill	billing_query	urgent	Billing	high	✓ Correct

Both emails were correctly classified and routed end-to-end; the L3 team received the CSV report as an email attachment.

07 — TEST EVIDENCE SCREENSHOTS



Logic App run — all steps passing



L3 team inbox – email received with CSV attachment

Excel interface showing a CSV report titled "subject" with columns: subject, sender, intent, sentiment, routing, team, priority, summary. The report contains two rows of data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1	subject	sender	intent	sentiment	routing	team	priority	summary																				
2	Mobile Internet Not Working Since Yesterday	galaxy.telecom.technical_support	urgent	Technical	high	Customer CUST-20481 reports mobile data stopped working since yesterday afternoon despite full 5G signal and having restarted the phone and reset network settings; voice/SMS work and they request urgent investigation and restoration.																						
3	Charged Twice for My Monthly Bill	galaxy.telecom.billing_query	urgent	Billing	high	Customer CUST-30245 reports being charged twice for the same monthly subscription (charges on 18 July and the following day), has had no response after raising the issue three days ago, and requests an investigation and refund of the duplicate pa																						

CSV report open in Excel – two triaged rows

Microsoft Azure

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Home > Storage center | Blob Storage > rgemailtriagedevaef4 | Containers

email-triage-output

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 📄 Rename 🔄 Acquire lease ⋮

email-triage-output

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 4 items

☐	Name	Last modified	Access tier	Blob type	Size
☐	📄 2026-07-23-triage-report.csv	23/07/2026, 23:32:42	Hot (Inferred)	Block blob	804 B
☐	📄 2026-07-24-triage-report.csv	24/07/2026, 13:36:19	Hot (Inferred)	Block blob	801 B
☐	📄 AQMkADAwATM3ZmYBLWI1OTMtO...	23/07/2026, 23:20:06	Hot (Inferred)	Block blob	420 B
☐	📄 AQMkADAwATM3ZmYBLWI1OTMtO...	23/07/2026, 23:20:05	Hot (Inferred)	Block blob	408 B

Blob Storage – email-triage-output container

Microsoft Azure

Search resources, services, and docs (G+)

Home > Function App

func-email-triage-dev

Function App

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Microsoft Defender for Cloud
- Events (preview)
- AI (preview)
- Resource visualizer
- Functions
 - App keys
 - Proxies
- Deployment
 - Add or remove favorites by pressing Ctrl+Shift+F

Browse Refresh Stop Restart Swap Get publish profile Reset publish profile Download app content

[Azure subscription 1](#) 512 MB

Subscription ID
a55387e7-ecf8-493a-8e9e-2e2467638f07

Tags ([edit](#))
Environment : dev Project : customer-service-ai

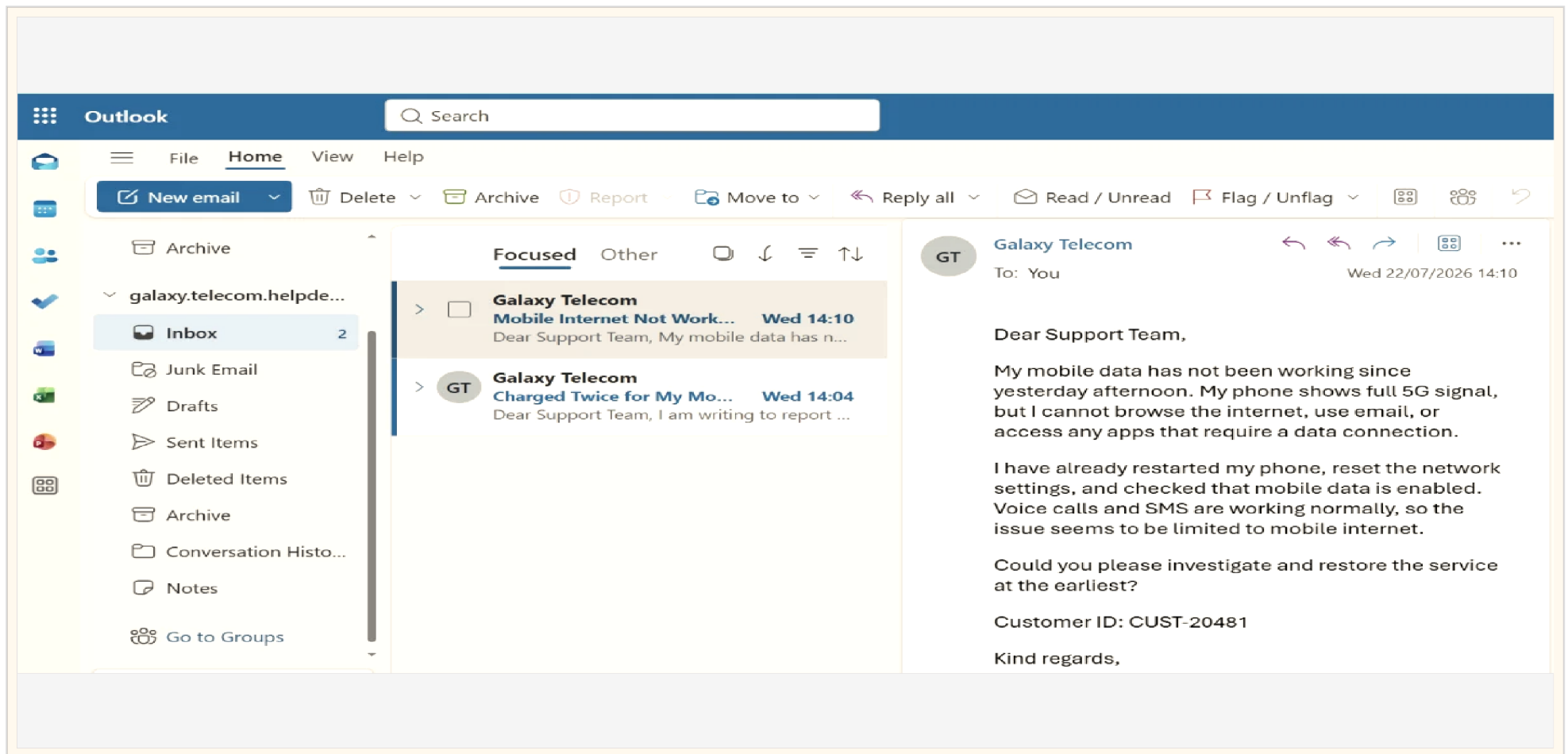
Functions Properties Notifications (0)

{ } Set up local environment Refresh

Filter by name...

Name	Trigger	Status	Monitor
analyse_email	HTTP	✓ Enabled	Invocations and more
send_report	HTTP	✓ Enabled	Invocations and more

Function App — analyse_email & send_report enabled



Helpdesk inbox – unread customer emails awaiting triage

08 — KEY LESSONS LEARNED

- Test each Function App directly before building the Logic App workflow around it
 - Flex Consumption uses a unique subdomain URL, not the standard `azurewebsites.net`
 - gpt-5-mini uses the Responses API — a different response structure from Chat Completions
 - Managed Identity cannot generate SAS tokens — an Access Key connection is required
 - Logic App string variables proved more reliable than array mapping for CSV building
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09 — KNOWN LIMITATIONS & NEXT STEPS

LIMITATIONS

- OAuth refresh token needs manual renewal on expiry
- SAS URI generation uses Access Key, not Managed Identity
- No deduplication of already-processed emails
- Plain-text email bodies only — HTML may introduce noise

NEXT STEPS

- Add deduplication via processed message ID tracking
 - Automate refresh token renewal on a schedule
 - Generate SAS tokens securely without Access Key
 - Add a Power BI dashboard over the CSV output; extend to CRM
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10 — ARCHITECTURE DECISIONS

DECISION	CHOICE	RATIONALE
Orchestration	Logic App	Separation of concerns; no code orchestration
AI Model	gpt-5-mini	Cost-effective, GA, strong reasoning capability
Auth	OAuth + Key Vault	Secure token storage with Managed Identity
Storage	Blob + lifecycle	Auto-archival after 7 days, deletion after 21 days
Deployment	Flex Consumption	Pay per execution, no idle cost
CI/CD	GitHub Actions + Azure CLI	Automated deployment on every push to main